

GLMP promotes EGFR-TKI resistance by activating autophagy and RhoA pathway in non-small cell lung cancer

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Experimental dataset evidence card

Document ID: DATASET-GSE298111

Category: experimental_datasets

Source entity: 03_experimental_datasets/DATASET-GSE298111.json

Dataset Identity

Dataset ID: GSE298111

Title: GLMP promotes EGFR-TKI resistance by activating autophagy and RhoA pathway in non-small cell lung cancer

Taxon: Homo sapiens

Sample count: 6

Selected reason: PC90R control versus si-GLMP replicates

Experimental Context

Assay context: resistance model; EGFR inhibition; osimertinib response

Sample accessions: GSM9008618; GSM9008621; GSM9008619; GSM9008622; GSM9008620; GSM9008623

Summary: Here, we report that the EGFR-TKI resistance mechanism is mediated by lysosome-related regulation. We established an osimertinib-resistant cell line, designated PC90R, derived from the parental PC9 cells. The overexpression of glycosylated lysosomal membrane protein (GLMP) in PC90R promotes resistance to Osimertinib. Mechanistically, GLMP could regulate the ubiquitination of RhoA and promote resistance by activating the epithelial-mesenchymal transition (EMT). Our findings provide a potential therapeutic strategy to overcome resistance to EGFR-TKIs.

Provenance

Provenance: derived_from_raw_layer

Privacy posture: cell-line or assay-level source selected

Raw source trace: 00_raw/_corpus/json/datasets/geo/GSE298111/geo_esummary.json;

00_raw/_corpus/txt/datasets/geo/GSE298111/series_soft.txt;

00_raw/_corpus/json/datasets/geo/GSE298111/source_record.json